

Low-Power BLE Sensor Device — Design & BOM

1 Features

Feature	Description
BLE Connectivity	DA14531 ultra-low power BLE SoC
6-Axis IMU	BMI270, accelerometer + gyroscope
Pressure Sensor	BMP280 or equivalent, locally available in Pakistan
Coin Cell Powered	CR2477 for 1-year operation
Ultra-Low Power	Target $< 50 \mu\text{A}$ average current
Minimalistic PCB	Bare-chip design, onboard antenna
External Connector	I ² C interface for pressure sensor
Waterproof Enclosure	IP67 rated protection
On/Off Button	Power gating for daily use

Table 1: Device Features

2 Bill of Materials & Pricing

Component	Part Number	Price (PKR)	Price (USD)
BLE MCU	Renesas DA14531	250	0.89
6-Axis IMU	Bosch BMI270	400	1.43
Pressure Sensor	Bosch BMP280	300	1.07
Battery	CR2477 Lithium Coin Cell	1000	3.57
Switch	Tactile Switch (SPST)	30	0.11
Connector	4-pin I ² C Connector	30	0.11
Passive Components	Crystals, Caps, Resistors	50	0.18
PCB & Assembly	JLC PCB Service	500	1.79
Total Cost		2560	9.15

Table 2: Component Pricing (Small Batch)

3 Technical Specifications

3.1 Power Consumption

- Target Average Current: $< 50 \mu\text{A}$ (event-driven + BLE interval $> 10\text{s}$)
- Deep Sleep Current: $\sim 0.5 \mu\text{A}$ (DA14531)
- IMU Low Power Mode: $\sim 450 \mu\text{A}$ (BMI270)

- Pressure Sensor: $\sim 3.6 \mu\text{A}$ at 1 Hz sampling

3.2 Battery Life

- CR2477 Capacity: 1000 mAh
- Estimated Life: 2+ years typical usage
- Guaranteed Life: 1 year with moderate daily use